

Two Walls, Two Rulers: What Context Compaction Actually Drops in a Real Agent Deployment

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Abstract

Long-horizon agents depend on context compaction. Prior work characterizes compaction using benchmarks, controller-generated trajectories, and — in at least one case — material drawn from real agent configurations and prompts. We report measurements from a **real, continuously used single-machine agent deployment: 48 session directories with at least one successful compaction, 203 successful compactions, and 41 recorded failure attempts** (snapshot 2026-09-19 00:3x local; the deployment keeps running, so every corpus-level number is paired with its snapshot). We report four findings, each with a measurement device that runs on ordinary session logs.

(1) Failures are three mechanisms, not one. Truncation (24), *no-shrink* (the summary is not smaller than what it shadows; 15), and a token-meter mismatch (2). Truncation and no-shrink require **opposite** remedies: one needs an output-budget strategy, the other needs a criterion for *refusing to compact* that span. Prior work measured an output floor; it did not name the failure mode that occurs below it.

(2) Summaries do not carry freshness — structurally, not by accident. The **Next Step** section is written *by the summarizer, at generation time*, and is rewritten only by the next compaction: between those two moments it reports as current a state the session has already moved past. **The section is not failing at its job; the job is undefined** — nothing in the summary format carries a timestamp or a validity condition. We show the consequence twice, and the second is the half that carries. **(a) It is acted on.** In the behavioural runs (§5), probes given one real summary **moved on a waiting condition the live session had already resolved** instead of reporting it, so the failure does not depend on how the question was phrased. **(b) Its form travels with it** — the half we file as **weak**: sixteen zero-history probes retrieved that same resolved sentence (**16/16**), a reading that proves little **by construction**, because the

material states a waiting condition and the question asks what it waits on, so a faithful reader answers correctly; we keep it only as the entry point to (a). What it still shows is narrow and worth one clause: a resolved statement travels in the **same surface form, with the same confidence, as a live one** — because **compaction rewrites; it does not timestamp**. (Summary text is paraphrased rather than quoted: not published, under our own data policy.)

(3) Two rulers, one phenomenon. Constraint items (the summary’s **## Boundaries** section) survive **verbatim** at **13%** after one compaction round and **0%** by round four. Judged **semantically** (three states: verbatim / reworded-but-present / gone) by ten zero-history judges, the same items survive at **91%**, **77%**, and **53%** ($k = 1, 2, 4$). One number per point invites the wrong question, so rounds 2 and 4 were re-judged by **two further independent sets of readers** (same packs, same rule, **102 cells each**). The result splits: the **shape reproduces, the deepest point does not** — $k = 2$ reads **76.8 / 76.8 / 78.6%**, $k = 4$ reads **53.3 / 45.7 / 45.7%** (majority **47.8%**), while at $k = 1$ four further readers of the same 56 cells span **82.1–91.1%**. §4 carries each reading, the interval on every point, and where the movement comes from — **pass-versus-readers, not reader noise**. Reporting **either ruler alone inverts the conclusion**, and no single number should be read as *the* survival rate.

(4) Substitution, not fabrication. In a task-completion setting, **twelve probes given the material without a retrieval card failed to select the operative “do not touch” constraint (12/12, three batches of four) and none of them marked the choice as a guess**; four probes given the same material **with** a card selected it (4/4). The card-versus-no-card contrast in the first batch is significant at this sample size (Fisher exact, two-sided, $p = 0.029$), and that significance **rests entirely on the zero**: had a single no-card probe selected the operative constraint, the two-sided p would rise to **0.14** and stop being significant (0.43 at 2/4, 1.00 at 3/4; `node tools/fisher.mjs`). They did not fabricate — they substituted a *different* constraint that genuinely appears in the material. **The two further batches of four no-card probes are reported separately rather than pooled**: in the second batch the frozen task text still pointed the windows at the apparatus document that records this probe’s answer (a defect we found afterwards and fixed), and in the third batch — with that pointer removed and the material moved out of the apparatus tree — the substituted targets **diversified** (two probes took the standing constraint that appears verbatim in the material, one took its own material path, one took the repository root). Across all three batches (12 no-card probes, 0 correct) the *zero* replicates: the operative constraint is never recovered. **Which** constraint gets substituted does not replicate, so we do not present “nearest-hand substitution” as the mechanism — **the invariant is the miss, not the substitute**. Across the first eight probes the number of searches was **0**; the four probes of the third batch were likewise **0/4** searches (scanned from their own session logs).

A human pass exists: one author, blind to the key, agrees with it more closely

than either machine pass (three-state **56.5%**, two-state **89.1%**) — our own prediction that human-machine agreement would be lower is not supported. **Three further readers, one from each of three different model families**, judged the same cells blind; where they differ from one another they split two-to-one and **never three ways**, and every disagreement with the key sits on the wording seam — **never** on whether a constraint survived. We do not claim a fix. Prior work shows that re-surfacing a plan does not restore task success; our positive results cover **state understanding**, not **behavioral success**.

Introduction

Long-horizon agents survive their own context window by compacting: an LLM summarizes the older part of a conversation, the summary replaces what it shadowed, and work continues. Every deployed system does some version of this, and the measurements that exist are mostly on benchmarks and controller-generated trajectories. This paper measures **one deployment that has been in continuous use** — a single-machine agent whose session logs record every compaction it has performed — and asks a narrower question than “does compaction help”: **what does it actually drop, and can the system tell?** **Material and method.** 48 session directories, 203 successful compactions and 41 recorded failure attempts at the §1.2 snapshot. The corpus is **live**, so every corpus-level number is quoted with the snapshot it was taken at; devices run on ordinary session logs, with no instrumentation of the compactor, and each is given an input whose answer we already knew: a fixpoint that must be 100%, a synthetic foreign corpus, and a positive control that must fail when the assumption behind it is wrong. **Contributions.** - **C1 (load-bearing).** Failures are **three** mechanisms, not one — truncation (24), *no-shrink* (15) and a token-meter mismatch (2) — and the first two need **opposite** remedies. Extended on the success side: the meter and the failure signal **disagree on about 9%** of successful compactions (20 of 231 — the live library on 2026-09-20, a different moment from the §1.2 snapshot; §2). - **C2 (a single structural observation).** A summary carries **no freshness**: the Next Step section is written at generation time and rewritten only by the next compaction, so an already-resolved state travels as current, and the section is not failing at its job — the job is undefined (§3). - **C3 (load-bearing).** Constraint survival depends entirely on the ruler: **13% → 5% → 0%** verbatim (normalized 5-gram) but **91 / 77 / 53%** semantically across independent zero-history judges — and the least stable of those points is the one we headline (§4). - **C4 (a direction, not a rate).** Given the material **without** a retrieval card, **12 of 12** probes failed to retrieve the operative constraint and none marked uncertainty; with a card, 4 of 4 did. **The miss replicates; the substitute does not** (§5). **What we do not do.** We claim **no fix**. Prior work shows that re-surfacing a plan does not restore task success, and our own material shows the same shape; we report

what we measured, where we measured it, and what would falsify it (§7, §8).

What we would try next, in order. (i) **Refuse to compact** a span whose summary is not smaller than what it shadows — the criterion *no-shrink* names, and the one place where nothing in the current design tells the operator to stop. (ii) **Carry validity on generated sections:** a **Next Step** that records when it was written and what would invalidate it, so a resolved state cannot present itself as current. (iii) **Make the retrieval card mandatory at task start** rather than optional — the arm C4 measures. We claim **no effect** for any of the three. Each is a direction our own readings point at, and each is falsifiable with the devices used in §2–§5.

What is about the step, and what is about this machine. The failure taxonomy and the timestampless section are statements about the compaction step **as specified here** — a compactor with an output budget, a size check, and no validity field — and any deployment whose logs record the same events can re-run the devices that produce them. Every *rate* we report is a rate about **one** deployment at a **named snapshot**, and each is quoted with its snapshot where it appears.

1. Setting and corpus

1.1 The deployment

The setting is a single-machine agent harness used continuously by one person for work, study, and personal assistance. Sessions are persisted as append-only logs, one directory per session. Agent turns, tool calls, and context events are recorded as typed events; context management (compaction) is a plug-in component of the harness, not a wrapper around it.

A **generation** is one successful compaction: a span of the session’s message history is replaced by a generated summary. The harness emits, per generation:

- a **compaction/summary** event carrying the summary text;
- a **compaction/receipt** event carrying a **deterministic retention audit**: which declared sections are missing from the summary prose, and which file paths and **UPPER_SNAKE** identifiers present in the shadowed original (or in the verbatim spill) are absent from it. **The receipt carries no token field** — the shadowed span’s estimate (**shadowedTokenCount**) is on **compaction/summary**, and the summary side has only the summarization call’s own usage, so the two numbers sit on **different metering channels** (§2, token-meter); (F1 receipt)
- the identity of the spans that were shadowed, including, in most generations, the *previous summary* itself.

A **failure attempt** is a **compaction/failure-class** event carrying the harness’s own error string. We count attempts, not sessions, and deduplicate where the same conversation was recorded under more than one directory.

1.2 Scale

Quantity	Value (snapshot 2026-09-19 00:3x)
Session directories with ≥ 1 successful compaction	48
Successful compactations	203
Recorded failure attempts	41 (in 6 sessions; de-duplicated lower bound 37)
Generations per session Concentration	median 2 , mean 4.2 — heavy-tailed two 29-generation sessions ($\approx 4\%$ of sessions) hold 58 of 203 generations ($\approx 29\%$)

The corpus is alive, and that is a methodological hazard. The same measurement pipeline four hours earlier (2026-09-18 17:2x) reported 46 / 192 / 41. A reader who recomputes and disagrees should check the snapshot before the method.

The deepest single session ran 29 generations across ten days: 29 summaries totalling 359,853 characters, of which **99.4% is text carried over from the summary that was itself consumed** — the compaction chain is a relay, not a funnel.

Evidence: `tools/compaction-census.mjs list|show|chain`; per-session aggregate columns in `data/sessions.md` (anonymized S01..S48; the identity mapping is not published).

1.3 Method

All four findings come from one pipeline: decode session logs $\rightarrow$ extract compaction events and their receipts $\rightarrow$ derive per-generation tables $\rightarrow$ apply a fixed criterion (written into the device header before use). The criterion is stated where it is applied, never in this text alone: changing a threshold means changing the device.

Use of AI tools. Reported under arXiv’s policy on generative AI language tools. An agent inside the deployment under study (**D**, a text-to-text generative tool) wrote the measurement devices, ran them, and drafted this report. The human author set the questions, reviewed each device against its inputs, checked every reported number against the device output, and **takes full responsibility for all contents, irrespective of how any part was generated**. Two consequences are recorded rather than left implicit: (i) the annotation key and the judges of §4/§8 come from **the same system under study** (§8, item 2); (ii) every work cited in §6 was checked against its own arXiv or publisher page (title, year, venue where one exists), and quoted sentences were matched against the source text — unchecked references are what the policy targets.

2. Finding 1 — failures are three mechanisms

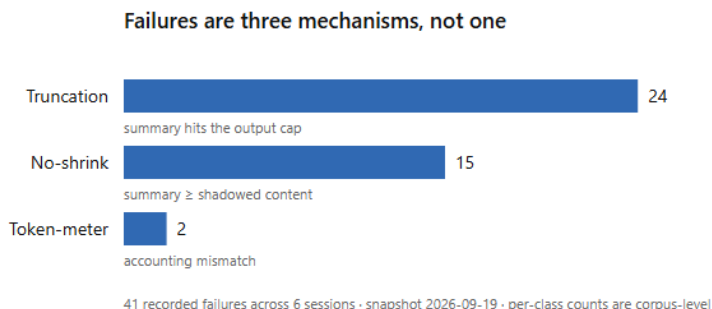


Figure 1. Failure mechanisms recorded in this deployment (n = 41 attempts): truncation 24, no-shrink 15, token-meter 2.

Mechanism	Verbatim error text	Count	Remedy direction
Truncation	summarization truncated at the token cap (incomplete checkpoint)	24	raise or segment the output budget
No-shrink	summary is not smaller than the shadowed content (4389 estimated framed tokens >= 3952)	15	a criterion for not compacting this span
Token-meter	(accounting mismatch on a different channel of the same session)	2	fix the measurement, not the compactor

The two dominant mechanisms are not variations of one disease. Truncation is an output-capacity failure — the summary was never finished. No-shrink is the opposite: the summary *was* finished and is not smaller than what it replaced, so compaction spent budget to save none. In the recorded case above, the framed summary exceeds the shadowed span by **437 estimated tokens**: the operation strictly increased context cost.

No-shrink is possible precisely because summarizers bound their own output. The literature has measured that floor (“output volume is essentially input-

and prompt-invariant across four backbones spanning 8B to 120B parameters”). What has not been named is the failure that occurs when the floor meets a span that is already small: the operator should stop, and nothing in the current design tells it to. In our corpus the two mechanisms co-occur in the same deployment and sometimes in the same session, which is why we report them as separate classes rather than as one failure rate.

Why this is not the intuitive failure. A successful compaction is recorded as a success: the summary replaces the span, the session continues, and nothing downstream compares the two. Where the two records disagree — the summarizer’s own metering beside the shadowed estimate (later in this section) — they are read by different consumers at different times, and neither is a gate. No-shrink is quiet for the opposite reason: the harness refuses, the session goes on, and the cost is visible only to a reader who counts `compaction/failure` events rather than `compaction/summary` ones.

Alternative explanations we could not rule out

- **The token estimate may be the artifact.** No-shrink is decided from the harness’s *estimated framed tokens*, not from measured usage. If that estimate is biased for short spans, some no-shrink events may be measurement artifacts rather than genuine non-reductions. We see no way to check it without instrumenting the compactor, which we did not do.
- **Attempts vs. outcomes.** We count failure *attempts*. We do not know how often an attempt was retried successfully, so 41 is not “41 losses”; it is 41 recorded refusals or errors.
- **Class boundary.** The third class (token-meter, n=2) was identified on a single session and may be an instance of the first two seen through a different channel.

Threat note. An output floor is known; the *no-shrink* failure mode below it is, to our knowledge, unnamed. **Failure signals catch only the extreme; on a second channel, one success in twelve is not smaller.** The no-shrink decision above is made on the harness’s own *estimated framed tokens*. The summarization call records, on the same event, its own metering next to the span it replaced (`usage.outputTokens` beside `shadowedTokenCount`). Over **231 successful compactions with both fields present** (live library, 2026-09-20 — a different moment from the §1.2 snapshot; zero missing values), **20 (8.7%) report provider output tokens at or above the shadowed estimate. All 20 sit in one of the four workspaces** in the library (122 successes there; $100 + 6 + 3$ in the other three combined, none net-negative) and their spans are small (1.1k–2.6k estimated tokens; library median ratio 0.034). Two different readers are piled into that sentence and must be separated: provider `outputTokens` measures **what the model generated**, not what was retained — for the worst cases a third ruler on the retained

summary text (characters $\div$ 4) puts it at ≈ 0.7 – 0.9 of the shadowed span while the provider reports 1.5 – $2.2\times$. So the defensible claim is narrower than “the operation was net-negative”: **the record of extreme failures and the record of metering disagree on about 9% of successful compactions**, which is the token-meter family of §2’s third class seen at scale rather than at $n = 2$. Positive control: the 15 recorded no-shrink refusals parse to $A \geq B$ in all 15 (median $A/B = 1.000$) — where the harness itself refused, the two channels agree. Reproduce with `node tools/net-shrink.mjs` (it prints its own control and refuses to run if the control fails).

3. Finding 2 — summaries do not carry freshness

Sixteen zero-history probes were given a single input: a real summary produced by this deployment’s compactor. Among the summary’s sections is **Next Step**, which at generation time recorded that the session was **waiting for the user to decide** a design question — a decision the live session had already made by the time the probes read the summary.

16 of 16 probes retrieved that sentence and reported it as the current blocking condition. In the live session, that decision had already been made and carried out — the very compaction under study was its execution. The summary was correct about the past and wrong about the present, and the probes had no instrument to tell the difference, because **compaction rewrites; it does not timestamp**.

A structural instance of the same problem: the **Next Step** section is written by the summarizer at generation time, so it is stale from the moment it is written until the next compaction rewrites it. The section is not failing at its job; the job is undefined.

Alternative explanations we could not rule out

- **Question priming.** The probes were asked what the material says is being waited on. A question phrased differently might elicit a hedged answer. (In the behavioural runs, §5, the same pattern appears without any such question: probes acted on the stale state rather than reporting it.)
- **One material family.** All sixteen probes read summaries from the same deployment and the same generation lineage. We have not shown that freshness loss is *general*; we have shown that at least one real, load-bearing statement came back dead, sixteen times out of sixteen.
- **No timestamp field.** The comparison is not “the summary forgot a timestamp”; the harness does not put one there. Whether adding one would help is untested.

Threat note. Related work measures *content loss* (rules erased, plans evicted, fidelity decay). Freshness is a different axis: an item can be perfectly preserved and simultaneously expired.

4. Finding 3 — two rulers, one phenomenon



Figure 2. Two rulers on the same materials: verbatim survival 13 / 5 / 0% and semantic survival 91 / 77 / 53% at $k = 1, 2, 4$; the band is the spread across independent judges.

We measured constraint survival two ways **on the same materials**, across 10 sessions whose summaries contain a **## Boundaries** section — the section the harness’s own summarizer instruction declares as *constraints and out-of-scope: what NOT to do* (`compaction-basic/src/sections.ts:33`), i.e. the deployment’s own name for its constraint section (**56** first-generation items; **46** in the $k = 4$ pool of the current device, **45** in the earlier extraction — see the note under the table):

k (compaction rounds)	verbatim (normalized, 5-gram ≥ 0.8)	semantic (“meaning still there”)
1	13% (7/56 · 95% CI 6–24%)	91% (51/56 · 95% CI 81–96%)
2	5%	77%
4	0% (0/46)*	53% (24/45)†

* *verbatim pool (46 bullets)* · † *pass-1 semantic cells (45)* — the note below says

why the two bases differ. Intervals are **Wilson 95% score intervals**, given where the counts are stated ($k = 1$); at $k = 2$ and $k = 4$ the dominant uncertainty is **reader spread**, reported in the next subsection.

Two denominators at $k = 4$ — and the difference is one judged cell, not one item. The verbatim ruler’s pool holds **46** bullets: the hand-sum of the generation-1 counts over the eight sessions that reach $k = 4$ (**7+5+9+5+6+4+5+5**, printed per session by `tools/constraint-survival.mjs`). The semantic reading tallies **45** cells at $k = 4$ in pass 1 and **46** after re-judging, and the single extra cell is visible in the tally device’s own output: in pass 1 that cell came back **unjudgeable** — **an answer outside the three-state vocabulary** (its later summary had kept only a place name for an item of personal information), so it was excluded from the pass-1 counts, and the re-judging pass answered it inside the vocabulary. The figure’s footer names both bases (“45 at $k = 4$ (46 re-judged)”). **Both bases give the same verbatim zero**, and the excluded cell is **not counted as a survival in either pass**. §8’s agreement figures (**52.2%**, $\kappa = 0.226$) are a **different quantity on a different base** — agreement with the key — not a second estimate of the survival rate.

The semantic column comes from ten zero-history judges, each reading one self-contained pack: the generation-1 constraint list, then the later summary. Each judge answered a three-state question — *verbatim* / *reworded but present* / *gone* — with a quoted basis, under an explicit “if you cannot find it, say **gone**” rule. The pack contained nothing else: no question about the study, no hint about the hypothesis.

Four conclusions:

1. **Compaction preserves meaning, not wording.** The gap is $\approx 7\times$ at the first round. A study reporting only the verbatim ruler would conclude that constraints are destroyed; a study reporting only the semantic ruler would conclude compaction is safe. Both would be reporting the same experiment.
2. **The two decay at different rates** (semantic: $91 \rightarrow 77 \rightarrow 53$, each point carrying an interval; verbatim: $13 \rightarrow 5 \rightarrow 4 \rightarrow 0$). “Verbatim death” is therefore not evidence of “constraint death” — and, conversely, a carrier designed to preserve *wording* addresses a different problem (a later window unable to quote the original) than a carrier designed to preserve *meaning*.
3. **A structural observation:** the **## Boundaries** section was never missing across generations (0 of 10 sessions lost the heading); what decays is the **items inside it**. The skeleton is stable; contents are rewritten each generation.
4. **Cross-device check.** An independently written device that measures strict verbatim inheritance over *all* bullet lines reports 4.6% average inheritance (2,131 distinct lines; only 16 lines survive ≥ 3 generations, covering 1.0% of the final generation’s characters). A stricter ruler must produce a smaller number than a looser one; observed $4.6\% \leq 13\%$, consistent.

Stability of the semantic column

One number per point invites the wrong question, so rounds 2 and 4 were re-judged by **two further independent sets of readers** (same packs, same rule, one cell per line, **102 cells each**): **k = 2: 76.8 / 76.8 / 78.6%** (original · re-read A · re-read B; **majority of three 78.6%**) and **k = 4: 53.3 / 45.7 / 45.7% (majority 47.8%)**. The **shape reproduces and the deepest point does not** — k = 2 moves **1.8** points between readings, **k = 4 moves 7.6**, and both fresh readings land below the original. At k = 1, four further readers of the same 56 cells span **82.1–91.1%** (the κ reading). Every interval is drawn in **Figure 2**. **Stability is not uniform across k, and the least stable point is the one we headline**: k = 1 spans **9.0 points** across four independent readings (82.1–91.1%), k = 2 moves **1.8**, k = 4 moves **7.6** — so the deepest point is *not* the least reliable one, and our **headline 91% carries the widest spread of the three**. That is a result about the ruler, not an apology about error bars: near “almost everything is still there”, independent readers disagree most about which near-misses count.

Where the movement comes from. The two re-reads agree with each other on **87 of 102 cells (85.3%)**; k = 2 80.4%, k = 4 91.3%), and of their 15 disagreements **10 fall on the boundary the curve does not use** (*verbatim* $\leftrightarrow$ *reworded*) and **5 on the one it does** (*still there* $\leftrightarrow$ *gone*). The point itself moves through a **different, larger set**: at k = 4 the **original pass and the two re-reads** disagree on **6 cells** (4 where the original said **still there** and both re-reads said **gone**; 2 the reverse; k = 2: 3 cells, net +1) $\Rightarrow$ the shift is **pass-versus-readers, not reader noise**. One of those six takes a nameable shape: the later summary **generalized the item until its object is no longer named** (the red line survives as “do not claim unlanded capability”, with the *specific* capability — and the layer distinction — gone), so both fresh readers called it **gone** where the original called it **reworded**.

Why the judges differ: one seam, not one disagreement

Three further readers — **three different model families**, same packs, same rule, blind to the key — labelled all 56 cells. Where they disagree with one another they split **two-to-one and never three ways** (14 of 56 cells), and **11 of those 14 sit on the verbatim $\leftrightarrow$ reworded edge**; collapsed to the two states the curve actually uses, the three agree on **53 of 56 cells (94.6%)**.

reader (A/B/C = three model families)	verbatim $\leftrightarrow$ reworded	reworded $\leftrightarrow$ gone	verbatim $\leftrightarrow$ gone	two-state vs key
A	14	7	0	87.5%
B	16	6	0	89.3%
C	17	6	0	89.3%

That is what the disagreement is: **not about whether a constraint survived, but about how much rewording still counts as verbatim** — the threshold our own rule never pinned down. The three ran one pack at a time, in order, inside a single session each, so unlike the pack-isolated passes their later judgments carry earlier context; we checked that separately and it points the other way (the pack-isolated passes lose **-35.0** and **-16.5** points between the first and second half of the material, these three **-19.0 / -8.5 / +1.9**), so the residual is not carry-over. They are reported as **their own arm** and are *not* folded into the judge decomposition above, whose counts depend on which judges are included.

A secondary effect we measured and do not claim

In a separate eight-probe comparison, the *order* of two constraint bullets in the same material moved selection between them (38% $\rightarrow$ 75% selecting the operative one when the two lines were swapped). The effect is directionally consistent across two rounds but **not statistically significant** (Fisher exact, two-sided, $p = 0.31$; a placebo swap left the other column unchanged, $p = 1.0$), and we stopped expanding the sample because the test statistic is non-monotonic in n and “expand until $p < 0.05$ ” is selection, not measurement. We report it as an observation about *which* constraint a window picks, not about whether it picks one at all.

Alternative explanations we could not rule out

- **The verbatim ruler penalizes paraphrase by construction.** That is its purpose; the semantic ruler exists because the verbatim number alone is not a statement about loss of meaning.
- **The judges are ours.** Thresholds, item splitting, and the three-state definition were written by the authors, and the judges were instantiated by the same system. **Two machine passes** have since been run — ten zero-history judges each, one pack per judge, each forbidden to read any other pack, covering all 56 cells — the second pass on a disambiguated version of the three-state rule, because pass 1 left a seam in our own wording (a sentence moved word-for-word into another section satisfied both verbatim and reworded). **We measure stability between passes, not against the key:** the key is one rater among five and the reference the survival curve is built on, not ground truth (§8 states the same limit). De-duplicated agreement with it was **52.2% (Cohen’s $\kappa = 0.226$)** in pass 1 and **54.3% ($\kappa = 0.261$)** in pass 2 — 25 then 24 disagreements out of 56; **between passes** the figures are **66–87.5% (three-state)** and **89–96% (two-state)** across the four independent runs, and those are the numbers we treat as the stability of the semantic column. **Clarifying the rule did not recover the disagreement**, so what remains is **judgment variance, not a rule defect. Neither figure is an inter-annotator number in the human sense:** same model, no history, and key and

passes share a model, so a human-versus-model figure can only be lower. Where the disagreement sits is still informative: the **verbatim** $\leftrightarrow$ **reworded** boundary carries the largest block (18 of 25, then 15 of 24) — a distinction the survival curve never uses. Collapsing to the two states the curve actually uses (*still there* vs *gone*) gives **87.5%** / $\kappa = \mathbf{0.470}$ in pass 1 but **83.9%** / $\kappa = \mathbf{0.319}$ in pass 2, so that boundary should be quoted as a **range (83–88%, κ 0.32–0.50 across two runs), not a point** — with 56 cells we make no significance claim between the runs. (Those two-state figures are **per cell**; de-duplicated they are **87.0%** / $\kappa = \mathbf{0.502}$ and **82.6%** / $\kappa = \mathbf{0.336}$. The three-state numbers earlier in this bullet are **de-duplicated** — we label each rather than quoting one ruler for both.) One further reading is worth stating because it points away from the judges: **the two passes agree with each other on 42 of 56 cells (75.0%), while each agrees with our key at 55–57%.**

- The outlier looked like the **key** — which was written while reading the whole session, whereas a judge sees only its pack. A **third pass** moves that reading and we now retract it: same material, same rule, but judged **one cell at a time**, so the reader never sees a cell’s siblings. With three windows the key stands alone on only **10 of 56** cells at three-state resolution and **3 of 56** on the two-state boundary, while the windows disagree *among themselves* on **24** and **6** cells respectively. **Two windows agreeing is therefore not consensus** — only a third window separates “agreement” from “these two happen to match”; with two passes we had been reading the former as the latter. The third pass also sits **closer to the key** (92.9% two-state) than either pack-level pass (87.5%, 83.9%), which is consistent with the key having been written item by item rather than pack by pack — but the contrast is **not significant** (McNemar exact, $p = 0.238$ / 0.302 three-state; 0.375 / 0.063 two-state), so we report a direction, not a result.
- A **fourth pass** closes the loop on the “our judges are the problem” worry: a **different vendor’s model** (Claude), given the same packs and the same rule, agrees with our key on **64.3%** of cells — 89.3% on the two-state boundary — and lands inside the same band as the machine passes. With four judges the key stands alone on only **2 of 56** boundary cells while the judges disagree among themselves on 8; **no pass is a systematic outlier** at any judge count, and the ordering that would say position matters more than model family (cell-level and cross-model raters agree with each other on 87.5% of cells, the highest pair in the matrix) is **not significant** (McNemar exact, $p \geq 0.267$). We report it as a hypothesis for a dedicated run, not as a result. A same-pack control (one pack judged twice by two independent zero-history windows under the identical clarified rule) came out **5 of 7** cells.
- A **four-judge decomposition** (key / pass 1 / pass 2 / pass 3 / cross-model pass, per cell) puts numbers on it — **Figure 3** draws these same

56 cells, aggregate counts only — and the numbers moved as judges were added: on the three-state labels all four judges agree with the key on **39.3%** of cells, all four agree with each other but differ from the key on **16.1%**, and the judges disagree *among themselves* on **44.6%** — the largest block. On the two states the curve actually uses, the same decomposition is **82.1% / 3.6% / 14.3%**: there the key stands alone on only **two** cells while the judges disagree with each other on **eight**. (Judge counts are not comparable to one another — more judges make consensus harder — so we read this as “no systematic outlier at any count”, not as a trend.) **With two passes we had read “the two agree” as consensus and called the key the outlier; neither the third judge nor the cross-model judge supports that reading** — which is why the sentence above is stated as a retraction. We then tested the most convenient explanation for that — that our key simply applied an older reading of the rule (the disambiguation was written *after* the key) — and it fails: among the 16 cells where our key stood alone against the first two passes, only **3 (19%)** have the original wording still present verbatim in the later summary, against **7 of 25 (28%)** among the agreed cells (criterion: normalized 5-gram containment ≥ 0.8 , a verbatim proxy, not a semantic test). The divergence is substantive, not an artefact of rule version. By that same proxy, **3 of those 16** cells have the wording still present, yet our key says **reworded** — and under a stricter criterion (the item’s full normalized text as a contiguous substring of the later summary) it is **1 of those 16**. Both counts are recorded as instrument defects in the key, not as properties of the corpus; the two criteria are monotone here (strict $\subseteq$ proxy, 3 of 56 vs 17 of 56, no reversals). Both figures are reported with the instrument defects attached, not as a property of the corpus, and the semantic column should still be read as a first measurement with wide error bars rather than a rate.

- **A human pass now exists, and it narrows that bar rather than widening it.** One of the authors — the human half of the deployment’s pair; the key and both machine passes were produced by the agent half — labelled all 56 cells after reading only the packs and the three-state rule, with no access to the key. Three-state agreement with the key is **57.1% per cell / 56.5% de-duplicated** ($\kappa = 0.282 / 0.291$), i.e. **at or above both machine passes**, so the machine figures are not an upper bound on human agreement: the prediction in our previous draft (“a human-versus-model figure can only be lower”) is **falsified**, and we record the retraction rather than the prediction. Her disagreements concentrate where the machine passes’ did: **19 of 24 sit on the verbatim $\leftrightarrow$ reworded edge** (machine passes: 18 of 25, then 15 of 24), against 5 on **reworded $\leftrightarrow$ gone**. On that edge she leans in **both** directions — 10 cells toward **reworded** where the key said **verbatim**, 9 the other way — where each machine pass leans one way (11 and 10 cells toward **verbatim**). On the two states the survival curve actually uses she is **the closest of the three**

raters to the key: 91.1% per cell / 89.1% de-duplicated ($\kappa = 0.568 / 0.556$), leaving 5 of 56 cells in dispute (machine passes: 7 and 9). Her pass ran under the **original** three-state rule — the packs’ own version, the one this bullet opens with a seam in — not the disambiguated rule. With **three raters, one of them an author**, this is still not a general inter-annotator study; it is the human tier this report was missing.

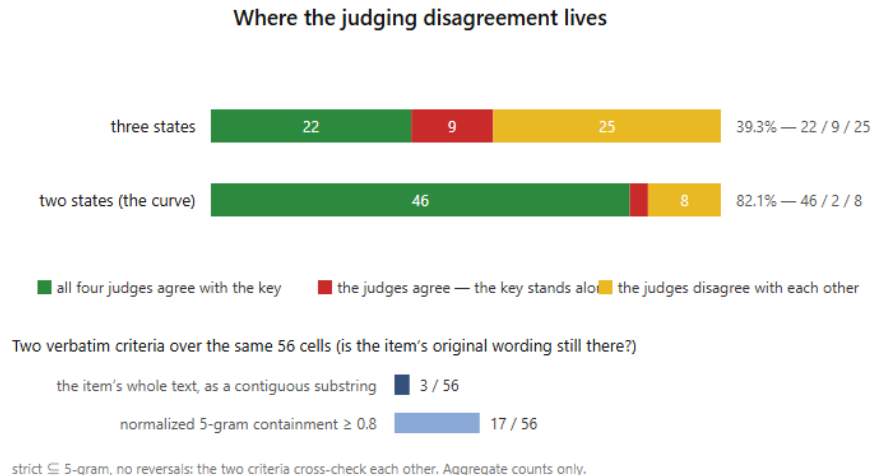


Figure 3. Where judging disagreement sits across 56 cells: 39.3% — 22 / 9 / 25 on the three-state labels, and 82.1% — 46 / 2 / 8 on the two states the curve uses.

- **Item granularity.** A compound constraint (“full authority, but report back”) can have one half present and one half gone; the three-state scheme collapses that to one label. One judge reported exactly this case unprompted.
- **Session selection.** The 10 sessions were those with ≥ 3 generations *and* a **## Boundaries** section. Older sessions in the corpus used a different section skeleton (8 sections, no **Boundaries**), and were excluded rather than mapped onto the new one.

Threat note. Compaction Cliff (arXiv:2608.22752) reports safety-rule recall of 53% after one round and 10% after five. ! **Correction (2026-09-19):** an earlier draft of this note called that corpus *synthetic* — **that description is one we cannot stand behind**, and the comparison does not need it. The reason our absolute levels are **not** comparable is **a different corpus scored under a different ruler** (their recall is a semantic criterion over material we did not sample), **not** a synthetic-versus-real contrast. Only the shape of decay is comparable. We also record an impossibility: AgingBench (arXiv:2605.26302) reports half-life in sessions, which requires a per-

session capability score — our logs contain no such quantity, so that ruler cannot be transplanted without changing what is measured.

5. Finding 4 — substitution, not fabrication

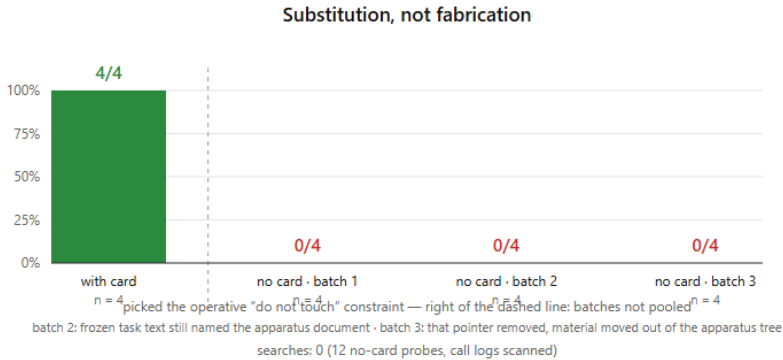


Figure 4. Behavioral probes: 4 / 4 windows retrieved the operative constraint when the material arrived with a retrieval card; without one, three independent batches returned 0 / 4 each (batches not pooled — §5).

Understanding is not doing. We therefore ran a second, behavioral measurement: probes receive the same frozen material as above, delivered either as **material without a retrieval card** or **material with a card** carrying one line that states the operative constraint and its reason. Each probe must produce a **next** pointer file plus a reason. Two columns are scored: which “do not touch” constraint it selected, and whether it searched.

Arm	n	selected the operative constraint	searches
with card	4	4/4	0
without card — batch 1	4	0/4	0
without card — batch 2 (<i>frozen task text still named the apparatus document that records the answer</i>)	4	0/4	—

Arm	n	selected the operative constraint	searches
without card — batch 3 (<i>that pointer removed; material moved out of the apparatus tree</i>)	4	0/4	0

The failures are informative. In batch 1, three probes selected a standing constraint that **does** appear verbatim in the material’s **## Boundaries** — one that is real, but not the one governing this task — and one selected a different file that also appears in the material. Counting occurrences in the material explains **that** batch: the operative constraint appears **0** times without the card and 3 times with it; the substituted constraint appears 3 times in both. **It does not explain batch 3:** same material, pointer removed, and the four probes took the material’s standing constraint twice, their own material path once, and the repository root once. Batches 2 and 3 are therefore **not pooled** with batch 1 — the three windows could see different things — and the mechanism claim is scoped accordingly: **the zero is the invariant; the substitute is not.**

This is **substitution**: not invention, but the capture of whatever constraint is nearest to hand — and, crucially, **without any marker of uncertainty**. The probes did not say “I am guessing”; they filled the slot. For safety-relevant behavior a confidently substituted constraint is worse than an acknowledged unknown, because nothing downstream flags it: the file gets written, the slot is filled, and the error is invisible until something acts on it.

We observed the same shape in three different material families earlier in the line: a material with no correct item present produced *substitution* (4/4 probes picked the nearest real constraint); a material that contained identifiers but no constraints produced an honest “cannot find a basis” (2/2); a material containing the real constraint plus a competing one produced *selection* (3/8 and 6/8 depending on order). The failure shape is a property of the material, not a single disease.

Across the first eight behavioral probes the number of searches was **0** (7–13 tool calls each, all within the supplied material plus writing their own output file); the four probes of batch 3 were also **0/4**. This is the first number we can attach to the design goal “continuity without searching.”

Alternative explanations we could not rule out

- **Task difficulty.** A harder task might have produced searching, or an explicit “I don’t know”. Our task was small on purpose: it isolates *which*

constraint is carried, not whether the window can do the work.

- **Zero-history probes are not working windows.** A real window has its own context, its own recent actions, and its own stake in the outcome. Nothing here shows how a working window behaves after a real compaction; the one observation we have from a working window ($n=1$, the authors’ own) is that it answered from context without reading files, and that its retrieved pointer was *stale* — consistent with §3 but not a distribution.
- **The card may be doing something else.** We assume the card works because it contains the constraint verbatim. An equally consistent account is that it works because it is *short*: a one-line carrier is easy to read. We did not run a short-but-wrong card.

Threat note. “Compaction erases constraints” and “plans are evicted early” are established. What we have not seen measured is the **shape of the failure after erasure** — a near neighbour substituted with full confidence — nor its behavioral consequence in a real deployment.

6. Related work

Nine recent works bear directly on this one. We list what each contributes, and what remains for a deployment study.

- **Compaction Cliff** (arXiv:2608.22752) — measures safety-rule recall decaying $53\% \rightarrow 10\%$ over five rounds on a production-style prompt, and proposes type-conditioned retention. Establishes that *content* decays and that retention policies should not be type-blind. We measure the same phenomenon on real sessions, in two rulers, and add the failure classes that occur *below* the output floor.
- **Governance Decay** (arXiv:2606.22528) — shows compaction silently erasing governance constraints, quantified by **ConstraintRot**, a **benchmark of self-contained long-horizon agent scenarios with deterministic violation grading** (plus a real-harness validation across production agent-framework context managers, §8), with **Constraint Pinning** as a remedy — a **training-free mitigation that quarantines the constraint from lossy compaction, re-injects it, and integrity-checks it across turns**. Our contribution is what happens in the *absence* of such a policy, on real material: substitution without a marker.
- **Lost in Compaction / COMPINT** (arXiv:2608.11242) — names **Session Constraints** (user instructions meant to bind the rest of a session) and measures their loss with a purpose-built suite across multi-turn chat, agentic trajectories and long-horizon research: current compactors retain **17% of injected constraints on average**, and **most perform worse than not compacting at all**. Retention varies with compactor, prompt, context length, constraint phrasing and injection location, so the loss is **systematic rather than a bad setting** — and they answer it on the

extractor side (an SC-aware extractor). This is the closest prior statement of the §4 phenomenon, and it sharpens what is ours: they **inject** constraints into constructed scenarios and score **retention**; we measure **real deployment logs** under **two rulers** (13% verbatim vs 91 / 77 / 53% semantic, §4) and add what happens **after** erasure — a near neighbour substituted with full confidence and no marker (§5) — which a retention rate cannot show. Their headline (“most compactors do worse than no compaction”) is also the second independent argument against reading our positive state results as a fix.

- **Plans Don’t Persist** (arXiv:2606.22953) — one-step plan-signal decay ($4.1\times$) and a compression stress test showing $56.7\% \rightarrow 22.0\%$ success after naive eviction, with no recovery from plan protection (20.7%, $p = 0.89$) or probe-gated re-surfacing (24.7%, $p = 0.67$), despite 6.1 re-insertions. This is the strongest argument against reading our positive state results as a fix.
- **Beyond Compaction / CWL** (arXiv:2606.11213) — typed, dependency-linked episodes with deterministic, **LLM-free** eviction. Establishes **deterministic, LLM-free eviction over agent-declared typed episodes** as a system design; we observe that in a deployment, deterministic artifacts exist but have **no automatic reader** (they are read 0 times unless something triggers a read).
- **Parallel Context Compaction** (arXiv:2605.23296) — summarization **output volume is essentially input- and prompt-invariant** (input grows $48\times$, output $\sim 3\times$; the model self-bounds its output) and **synchronous compaction dominates wall time at low thresholds** (up to 62% of end-to-end execution). Establishes the output floor we cite in §2.
- **Rate-Distortion survey** (arXiv:2607.08032) — unifies compression approaches **under one rate-distortion objective and promotes reversibility and query-conditioning**. Its (P-fid) **fidelity profile** axis is adjacent to what our two rulers separate: preserving *wording* and preserving *meaning* are different targets — and its own (P-rev) reversibility axis is about re-derivation on demand, not about wording.
- **Aging / AgingBench** (arXiv:2605.26302) — names deployment-time agent aging and measures it with a synthetic generator plus longitudinal statistics (half-life, decay slope), noting that production telemetry is an open frontier. This paper is a data point on that frontier — and a demonstration that one of its rulers cannot be transplanted to logs that lack per-session capability scores.
- **SHARP** (arXiv:2605.06822; policy evolution under a validation gate) — in a different domain, names the failure we measure in §5: unbounded text rewrites can **silently overwrite previously correct behavior** (*policy degeneration*) in ways a validation gate *cannot fully catch*. Its controlled ablation is the cleanest available statement that structure in the **mutation operator** — not model or budget — decides the outcome: same validation gate and same evolution budget, only rule-editing replaced by free-form rewriting, and **on GPT-4o-mini** performance falls **below**

the no-adaptation baseline (Sharpe $+2.45 \rightarrow -0.84$; total return $+33.2\% \rightarrow -12.1\%$, against **+7.3% static**). On the second backbone the same ablation lands at **+6.0%** versus a static baseline of **-6.4%** — above static, still well below full adaptation ($+22.9\%$) — so the *size* of the degradation is backbone-dependent while the free-form variant trails full adaptation on both. The same paper reports the shape we measure: the free-form variant “gradually shifts ... into flowing prose whose thresholds drift without any traceable link to specific error days”. We measure the same family on real deployment logs (substitution without a marker, §5) rather than through a downstream reward.

One axis we probed three times, because it is the axis we would most like to have: *which harness*. Across the **eight works in that scan** (the ninth entry above was added after that scan — its own axes are compactor, prompt, context length, constraint phrasing and injection location, again **not** implementation), the comparison axes are model (5), task (5), run (3), session (3), benchmark / domain / layer (2 each), backbone / scenario / seed / strategy (1 each) — *implementation* is none of them (device, spans and controls in `report/prior-art-cross-impl.md`). The two nearest runs vary the container while holding the compactor fixed: Governance Decay §8 injects the *same* LLM-written compacted summary into LangGraph, LangMem, AutoGen’s `BufferedChatCompletionContext` and an OpenAI Agents SDK Runner, and labels its SDK case “an SDK integration check rather than a claim about SDK-native compaction”; AgingBench does run ReAct, OpenHands and Claude Code — including production CLI agents with recompaction events — but scores the **aging curve**, with the memory policy and compaction prompt supplied by its runner (App. F.3). The rate-distortion survey’s own §13.2 names the same gap we address: no suite checks “whether a scheme knows what it dropped, whether it can get it back, and whether its confidence reflects what it kept”. So our claim is a first measurement **on an unoccupied axis**, not a cross-implementation comparison: the harness that produced our results is one. What we can say about the devices is narrower and stronger than “portable”: the two things a device must know about a corpus — *how logs are stored* and *what the events are called* — are isolated in one editable table (`tools/foreign-log-adapter.mjs`), and the device now runs on **two harnesses’ native logs**, verified against independent counts on both (see §9 and `report/cross-harness.md`). **Our own measuring side has since crossed one implementation boundary in the other direction:** pointed at **198 native transcripts from a different harness**, the adapter paired **29 of 29** generations at id level and the independent counts agreed (§9) — the axis is empty in the literature while being *traversable* by the apparatus.

What a head-to-head would require. We deliberately do not place our numbers beside anyone else’s, and the reason is not modesty: a numeric comparison is only meaningful when three things match — the same material, the same **frozen** criterion, and the same round k . Two of our results are portable in that sense: the verbatim curve is defined by normalized 5-gram containment

and the semantic curve by a written three-state rule, both fixed before use, so a reader can re-run those criteria on their own summaries and obtain a comparable number — that is the comparison we would rather invite. The third result is **not** portable: it measures whether a retrieval card changes what a window does, which is a property of one deployment’s apparatus rather than of summaries. The one cross-material reading we do report (§9) compares *material and compaction regime*, not implementations. And where a prior work states a same-shaped claim on a different ruler — Compaction Cliff’s 53% $\rightarrow$ 10% recall of safety rules — we cite it as a claim (threat note under §1) and we do not subtract our curve from theirs.

7. What we do not claim

Claim we would like to make	Already established by
Compaction erases constraints and rules	Compaction Cliff; Governance Decay (with a benchmark)
Plans and pointers are the first to decay	Plans Don’t Persist
Reversible memory beats irreversible memory	Rate-Distortion survey
Retain by pinning (Governance Decay) / by agent-declared structure (CWL) rather than by detecting stability	Governance Decay (pinning); Compaction Cliff (per-type); Beyond Compaction (agent-declared episodes)
Models largely ignore compaction length instructions	Parallel Compaction
Deterministic, LLM-free eviction over agent-declared episodes can replace LLM rewriting	Beyond Compaction (CWL)
Unbounded free-text rewriting silently degrades previously-correct behavior, and a validation gate cannot fully catch it	SHARP (arXiv:2605.06822, <i>policy degeneration</i> ; controlled A2 ablation)

We also do **not** claim that a retrieval card fixes behavior. The strongest available evidence points the other way (§6).

8. Limitations

1. **One deployment, one person.** 48 sessions from a single machine are a case study, not a distribution. Treat every rate here as an existence proof plus a first measurement.
2. **Self-judged.** All thresholds were fixed by us before use; the semantic column was produced by zero-history judges instantiated by the same

system. **Two machine passes** now exist (ten zero-history judges each, one pack per judge, all 56 cells): de-duplicated agreement **52.2%** / $\kappa = 0.226$, and **54.3%** / $\kappa = 0.261$ after we disambiguated our own three-state rule — the rule fix moved it by ~ 2 points, so the residual is **judgment variance**, not the wording. On the two states the survival curve uses, the two passes give **87.5%** / $\kappa = 0.470$ and **83.9%** / $\kappa = 0.319$ (per cell; de-duplicated 87.0% / $\kappa = 0.502$ and 82.6% / $\kappa = 0.336$): quote the range.

- The two passes agree with **each other** on 75.0% of cells while each agrees with the key at 55–57%; a **third pass**, on the same material but judged one cell at a time, shows that two windows agreeing is **not** consensus — with three windows the key stands alone on **3 of 56** cells on the two-state boundary while the windows disagree among themselves on **6**, so we no longer claim the key is the outlier. The third pass is nearest to the key (92.9% two-state) but **not significantly** so (McNemar exact, $p = 0.063$ –0.38).
 - A **cross-model pass** (a different vendor’s model, same packs and rule) agrees with the key at 64.3% three-state / 89.3% two-state and is not an outlier either; with four judges the key stands alone on **2 of 56** boundary cells — but judge counts are not comparable to one another (more judges make consensus harder), so we read that as “no systematic outlier at any count”, not as a trend. Both are *repeat-run stability* figures, not human agreement, and 56 cells support no significance claim between runs.
 - A **human pass now exists**: one author, blind to the key, labelled all 56 cells under the original three-state rule — de-duplicated agreement **56.5%** / $\kappa = 0.291$, i.e. **at or above both machine passes** (our earlier prediction that a human-versus-model figure must be lower is **falsified**, and is recorded as a retraction), and on the two states the curve uses **89.1%** / $\kappa = 0.556$, the closest of the three raters to the key, with her disagreements on the same **verbatim** $\leftrightarrow$ **reworded** edge (**19 of 24**). What is still missing is not a human but a **non-author** human: the inter-annotator number remains $n = 1$, and the key is written by the agent half of the same pair.
 - **Three further readers, one from each of three different model families** (blind to the key, one pack at a time in order, one session each) were added after this list was first written; they are reported as **their own arm** in §4 and are **not** folded into the judge counts above, because those counts depend on which judges are included. Their disagreements with the key sit **entirely off the two-state boundary** — **0** cells at **verbatim** $\leftrightarrow$ **gone** against **14 / 16 / 17** at **verbatim** $\leftrightarrow$ **reworded** — which is the same seam every other pass falls on.
3. **Small n — intervals where counts exist, reader spread everywhere else.** Survival curves: 10 sessions / 56 items, with **Wilson 95% intervals** at $k = 1$ (7/56 $\rightarrow$ 6–24%, 51/56 $\rightarrow$ 81–96%); at $k = 2$ and $k = 4$ the

dominant uncertainty is **reader spread**, which we report per point rather than as an interval. Behavioral: 4 probes per arm in the first batch, and 12 no-card probes across three batches that are **not pooled** (their windows could see different things) — at $n = 4$ per batch we quote no interval at all.

4. **Transplanted rulers.** Where a published ruler could not be transplanted (AgingBench’s half-life), we say so rather than substituting a different quantity silently.
5. **No success metric.** We did not measure whether tasks succeeded more often; that requires a task suite we do not have.
6. **Failure counts are attempts.** See §2.
7. **The probes were not fully blind to the apparatus — and we can say exactly which channel stayed open.** Two channels could hand a window the answer: the frozen task text of one batch named the apparatus document that records the probe’s answer, and every later batch carried the **author’s own working notes** — which mention the discriminator — in its injected context. We scanned each batch’s session log for both channels rather than assuming blindness: batch 1 shows no trace of either; batch 2 shows traces of the document channel; batch 3 (materials and prompt moved out of the apparatus tree) closed the document channel at **0 traces**, while the note channel remained and is **quantified**: three string families unique to our own notes appear in all four probes’ injected context at identical counts (4, 13 and 6), which is what tells us they come from injection and not from the probes’ own writing. We then edited the note that carried the discriminators out of the line’s own injected map (re-count for those families in that file: 0), but the same wording also lives in the author’s durable memory, and **injection is not configurable per run from inside the deployment** — so a fully blind condition needs a *different* deployment or agent, not another batch, and we report that as the residual rather than as an achieved control. The zero itself replicated under all three conditions — normal prompt, non-blind prompt, and prompt-with-pointer-removed — which is why we call it the invariant, and the apparatus defect found this way (a frozen task text that still pointed at the apparatus document) is recorded in the line’s own notes.

9. Reproducibility

Every number above is produced by one of six devices in `tools/`, all of which run on ordinary session logs and print the criterion they apply:

Device	Measures	Input
<code>compaction-census.mjs</code>	success/failure per session, shadowed tokens, receipts	session logs

Device	Measures	Input
<code>constraint-survival.mjs</code>	verbatim survival curve	census dump
<code>constraint-survival-pack.mjs</code>	builds self-contained judgment packs	census dump
<code>semantic-survival-tally.mjs</code>	aggregates three-state judgments	judge answers
<code>probe-toolcalls.mjs</code>	tool calls and search count from a probe's own log	session log
<code>scan-prior-art-axes.mjs</code>	which comparison axes a body of prior work uses, with positive/negative controls (§6)	a directory of arXiv HTML

All six were executed inside the published tree before release and reproduce the numbers in this report (`README.md` records the check). Thresholds live in the device headers, not in this text. The protocol for the probe runs — material/question/key discipline, the three scoring columns, the search-counting rule with its two traps, and the registration rules — is in `tools/PROTOCOL.md`. The §6 axis tally, its verbatim spans and its controls are in `report/prior-art-cross-impl.md`.

A portability test, and what it caught. `tools/foreign-log-adapter.mjs` isolates the two things a log-reading device must know about a corpus — *container/layout* (here: multi-frame zstd, gzip JSONL, plain JSONL) and *event vocabulary* (a rename table plus field aliases) — into tables, so pointing a device at another harness means editing a table rather than the logic. `tools/make-foreign-fixture.mjs` then generates a **synthetic log in a deliberately foreign shape** (plain/gzip JSONL, events named `ctx.compact` / `ctx.compacted` / `ctx.compact_receipt`, fields named `tokensShadowed` / `seqs` / `cid`) and states the numbers the census must print. Running it end to end found a defect that had nothing to do with the second harness:

The census matched each `compaction/start` to its summary by **seq equality**. On this corpus that assumption never holds (`start` seq 129,373 / `summary` seq 129,374), so the section headed “*triggered a start but produced no summary*” was listing **every** start — all 36 in the deepest session, of which 30 had in fact produced summaries. The correct key is `compactionId`, which all four event kinds carry. After the fix the same session reports **6** unpaired starts (36 - 30) and the one genuinely unclosed start is unchanged.

Two things generalize. First, **a detector that always fires looks like a finding**: this section had been printed, read, and quoted in our own notes as a

real property of the corpus. Second, the cheap way to find that class of error is to **run the device on an input whose expected output you know**, which is exactly what a synthetic fixture gives you — the same discipline as the empty-input guard, applied to the *shape* of the input rather than its presence.

And then a real second harness appeared. After the synthetic fixture, the same adapter was pointed at a genuinely foreign corpus that happened to exist on the same machine: **198 native transcripts from a different agent harness** (plain JSONL, so the container seam matched immediately). Compaction is not named as an event there; it is marked by a boolean field and a subtype. Adding two predicate rules to the adapter table — `system+subtype:'compact_boundary' → the accounting record`, `user+isCompactSummary:true → the summary record` — was enough for the census to run on it. On the largest transcript (**11,241 records, 29.4 MB**) the census reports **29 compactions**, matching an **independent count of the 29 boundary records**, and a shadowed-token total of **4,905,460**, matching an independent sum of the same field. Pairing between the two record kinds was established by an **id, not by adjacency** (`summary.parentUuid == boundary.uuid`, holding **29/29**) — deliberately, because assuming an identity from position is the defect reported above.

Three limits we state rather than bury: (i) the token field we map to “shadowed” is the *pre-compaction context size*, so it is an **upper bound**, not the same quantity our dsh numbers report; (ii) that corpus records **no failures**, so 0 failures there means *not visible*, not *none happened*; (iii) the **§4 ruler did not transfer** — see below. Our claim therefore remains what §6 says: a first measurement on an unoccupied axis, with a device demonstrated on two harnesses and real results from one. The foreign transcripts stay out of the artifact entirely: no upstream text is quoted, and the readings are counts, field names and rates.

What did not transfer, and what did instead. Our constraint ruler keys on a **named section** of the summary. The second harness’s summaries contain **no ## heading at all** — 0 of 29 generations — so that ruler extracts nothing there, and we report *zero items found* rather than a percentage (an empty extraction is not a measurement of zero). A **structure-free variant** of the same survival question keeps the identical containment criterion and changes exactly one thing — items are bullet lines *anywhere* in the summary — and runs on both harnesses. Its reading is not harness-shaped: the two long sessions, one per harness, keep **4%** and **5%** of their first generation’s items verbatim after one round, while a dsh **burst-regime** session keeps **71%**. In this sample the variable that separates them is the session’s compaction regime, not the harness. Limits: the item class is bullet-level items, **not** constraints; one session per harness; regimes unmatched; $k = 0$ is 100% by construction and is used as the device’s own fixpoint check. Details and the full 12-session table: `report/cross-harness.md`.

The invariants the census never checked. The same fixture carries a second corpus that is *deliberately broken*, plus a look-alike negative control,

and `tools/audit-census-invariants.mjs` turns the census’s implicit counting assumptions into four stated invariants: (i) one summary per `compactionId` (otherwise a retry is counted as a second success), (ii) every summary carries a shadowed-token count (otherwise the shadowed total is silently low), (iii) checkpoint `seqs` are unique (otherwise the carried-over-text numerator double-counts), (iv) every `start` carries the pairing key (the ERR-047 failure, inverted). On this corpus all four **hold** — 660 log files, 258 summaries, 303 starts, 258 checkpoints (**snapshot 2026-09-22 16:44**) — and on the decoy corpus the audit goes red on exactly the three invariants it should, with the offending records named. **These counts are not the headline counts, and the difference is the point:** they come from a *later* run of the same store, which keeps growing (the headline snapshot at 2026-09-19 00:3x counted 203 compactions, this run counts 258 summaries); every corpus-level number in this paper is paired with the moment it was read, and this is what that discipline looks like when the corpus is alive. The look-alike control is the one that tests *membership*: a message from a different plugin whose text is shaped like a summary is **not** counted as a checkpoint (criterion is the `source.plugin` field, not the content’s shape).

Three defects came out of our own tooling, and all three answer a different question without erroring: the census read its workspace filter from the **fourth** positional slot, so the intuitive `list <scope>` printed the whole library — formatted exactly like a scoped answer (it now takes `--scope` and warns when an argument is ignored); a device pointed at the wrong input directory printed an **empty table and exited 0**; and the front page had no copy-pasteable command, with paths that only worked on our machine. What found them was a cold read of this artifact as a stranger would read it, following the quickstart literally; their generalised forms are items (iv)–(vi) of the checklist below. We record the family because it is the failure mode of exactly this kind of work: the numbers are reproducible *in the authors’ environment* while the artifact silently fails elsewhere. **Data.** We publish code and **aggregate, anonymized** columns (`data/sessions.md`, one row per session). We do **not** publish session transcripts or summary texts: they contain one person’s private work and personal context, and our policy (`DATA-POLICY.md`) treats that as non-publishable regardless of research value.

A checklist for anyone repeating this. (i) Fix a snapshot and print it beside every corpus number. (ii) Measure the noise floor before interpreting an arm difference. (iii) Pick the discriminating column before the run (`[guess]` beats totals in our experience) and keep the frozen key untouched when it turns out too narrow — add a second key. (iv) Verify tool-argument key names against a dumped raw event; they are harness-version dependent. (v) When a device is copied into a publishable tree, de-identify it and re-run it there before claiming it works. (vi) **Give every device an input whose answer you already know.** We found three defects this way in one night, and not one of them was about the data: an empty input that printed a plausible table and exited 0; a pair of event kinds matched by position instead of by id, so a section headed “triggered but produced no summary” listed *every* attempt; and a filter read from

the wrong argument slot, so a scoped query silently answered the whole-corpus question. (vii) **Also give it an input you know is broken.** A detector whose predicate is always true prints a list that looks like a finding; a corpus that deliberately violates each invariant is the cheapest way to prove your checker is not permanently green. (viii) **Assert a fixpoint.** Our first version of one device produced ~0% on both harnesses — plausible as a result, and wrong, because a callback signature bug made every item after the first incomparable. Its saving grace was a cell that must be 100% by construction (a baseline compared against its own generation), now an assertion that fails loudly. Three cheap questions, each one asked of the device rather than of the world: *empty?* *known-good?* *known-bad?* *fixpoint?*

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Appendix A — per-session aggregate table

See `data/sessions.md` (48 rows; anonymized identifiers; columns: successful compactations, failed attempts, shadowed tokens, log size). Per-session failure *classification* is not available: it would require re-decoding every log, and we state the aggregate-only result rather than implying precision.

Appendix B — judgment protocol (semantic ruler)

Each pack contains: (a) the numbered constraint list from the first generation; (b) the full text of the summary after k rounds; (c) the three-state rule with an explicit fallback (“if not found, write **gone**”). Judges are zero-history: they read only their own pack, are forbidden from searching anything else, and must quote a basis for each verdict. Aggregated by `semantic-survival-tally.mjs`. The machine second pass reused these packs unchanged. **The human pass has now been run on the same packs** — one author, no access to the key, all 56 cells — under the **original** three-state rule this pack carries, not the disambiguated rule written afterwards (§8 reports the second pass under that one). Its agreement figures are in §4 and §8; the filled table and its scoring output stay in the annotation workspace and are not part of this tree.